OpenOmicsBench Version 1 Release Review

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Release position

The local version 1 candidate passes its scientific and repository preflight. It contains 12 certified biological objects from seven Expression Atlas studies. Publication remains paused until an independent person completes the quickstart, Zenodo supplies the exact version DOI, and the uploaded archive is checked against the final file inventory.

Measure	Result
Software version	1.0.0
Certified biological objects	12
Source studies	7
Redistributable pockets	12
Organisms	Homo sapiens, Mus musculus, Arabidopsis thaliana
Automated tests	45 passed
Collection overlap audit	PASS
Local preflight	PASS
Publication decision	NO-GO

Scope

Version 1 is a bulk RNA-seq count-matrix collection for testing analysis software, teaching reproducible workflows and developing methods on compact, traceable data. It does not claim read-level quality control, alignment validation, clinical suitability or complete coverage of each source experiment.

Ownership and licences

GitHub development is attributed to vxxqv. The Zenodo creator is Vivaan Patni. Project code is Apache-2.0. Copyrightable Expression Atlas material is included under CC BY 4.0 with provider credit. Metadata has a separate CC BY 4.0 licence file.

The biological collection

Each row is a separate benchmark object with its own contrast, ordered samples, pocket, expected result and provenance. Several objects may use one factorial source study, but their sample subset or statistical comparison differs.

Object	**Source**	**Organism**	**Samples**	**Pocket**	**Design**
rnaseq-002	E-MTAB-8572	Homo sapiens	10	2,000	balanced genetic knockout xenograft
rnaseq-003	E-MTAB-6866	Arabidopsis thaliana	6	4,000	balanced plant genetic knockout
rnaseq-004	E-GEOD-33979	Mus musculus	6	4,000	balanced mouse genetic knockout
rnaseq-005	E-MTAB-567	Homo sapiens	28	8,000	paired tumour and adjacent tissue
rnaseq-006	E-MTAB-8845	Arabidopsis thaliana	12	500	factorial plant infection response
rnaseq-007	E-MTAB-9206	Homo sapiens	6	8,000	balanced cell-line RNA interference
rnaseq-008	E-MTAB-10322	Mus musculus	9	8,000	imbalanced disease comparison
rnaseq-009	E-MTAB-8845	Arabidopsis thaliana	12	500	factorial plant genotype response
rnaseq-010	E-MTAB-8845	Arabidopsis thaliana	6	500	stratified wild-type infection response
rnaseq-011	E-MTAB-8845	Arabidopsis thaliana	6	500	stratified mutant infection response
rnaseq-012	E-MTAB-10322	Mus musculus	6	8,000	balanced Dmd-mdx genetic comparison
rnaseq-013	E-MTAB-10322	Mus musculus	6	8,000	balanced Dmd-mdx-beta-geo genetic comparison

Five annotation profiles are represented: Ensembl 95 GRCh38, Ensembl 97 GRCm38, Ensembl 107 GRCh38, Ensembl Genomes 44 TAIR10 and Ensembl Genomes 51 TAIR10. Pocket sizes range from 500 to 8,000 genes.

Selection and scientific validation

The source count matrix and experiment design are staged together. Sample names must agree, counts must be non-negative integers, repeated run columns must be exactly equal, and the declared design must retain enough information for the intended comparison. Candidate pocket sizes are fixed before scoring. The smallest candidate meeting all four minimums is selected.

Measure	Minimum required	Lowest v1 result	Interpretation
Effect-rank correlation	0.90	0.979	Ranks of shared log2 fold changes
Top-feature overlap	0.60	0.724	Jaccard overlap of the 50 largest effects
Sample-distance correlation	0.90	0.916	Preservation of log2(CPM + 1) sample geometry
Effect-sign agreement	0.90	0.948	Direction agreement where source effect is at least 0.1

All 48 object-level metric results pass. Values range from 0.724 to 1.000. Exact values remain in each expected/validation.json and expected/deseq2.json record, and in catalog/collection.json and catalog/collection.tsv.

Independent baseline

Every object was compared with its full selected-sample matrix under R 4.5.3 and DESeq2 1.50.2. The evidence records pin the input hashes, workflow hash, model terms, contrast, thresholds and measured values. This comparison confirms pocket fidelity; it does not repeat read alignment or feature counting.

Reference gate

Official Ensembl genome and annotation checksums are recorded, downloaded GTF files pass archive and hash checks, and every pocket identifier occurs in the matching annotation. E-MTAB-7126 was rejected because its current matrix did not match the annotation stated by the source. E-MTAB-5477 was rejected because no tested pocket met all four minimums.

Provenance, rights and overlap

Each distributed object carries a complete manifest with byte counts and SHA-256 hashes, a dated rights record, provider attribution, a reference record and a transformation record. The transformation record identifies the source files, selected columns, contrast, candidate grid, software versions, workflow hashes and commits.

Overlap review

The audit finds 12 unique object keys and 12 unique titles. No sample identifier is shared across source accessions. Shared samples occur only among declared contrasts or strata from E-MTAB-8845 and E-MTAB-10322. The only identical source matrix is used by rnaseq-006 and rnaseq-009, which apply different blocked contrasts to the same 12 samples.

Reuse reviewed	Finding	Classification
Rights records	One identical provider licence record across 12 objects	Expected shared legal evidence
Source count matrices	One identical matrix in rnaseq-006 and rnaseq-009	Same study, distinct contrasts
Long prose	Two citation groups repeat within their source study	Required source citation
Cross-study samples	None	No conflict

The audit treats required citations and legal records separately from accidental duplication. Object-specific descriptions, limitations and rights statements remain distinct.

Source limits

The collection begins with archive count matrices. Strandedness is recorded as unknown when the source does not state it. Users who need complete experiments, alternate contrasts or read-level evidence should follow the source accession and citation in the object card.

Software and reproducibility

The command line lists objects, shows manifests, retrieves declared tiers, validates data, reports provenance, checks the cache and diagnoses the local installation. A user does not need to inspect Python code to follow the normal path.

Check	Coverage
Manifest contract	Strict schema, round-trip test, unknown-field rejection and generated-schema drift check
Files	Declared paths, roles, byte counts, hashes and protection against path traversal
Counts and design	Integer values, sample order, duplicate labels, design rank, declared normalization and thresholds
Retrieval	Size and SHA-256 before atomic replacement; interrupted and corrupt transfer tests
Paired reads	Mate identity and selected-pair count regression test
Generated results	Fixture, catalog, figure, overlap report and release preflight

The 45-test suite passes locally. Continuous integration uses Python 3.12.14, has a five-minute limit and runs on every push, pull request and tag. It rebuilds the fixture and catalog, checks generated-file drift, validates all 14 registered records, runs the overlap audit and executes the release preflight.

Environment record

runtime/environment-lock.json records the tested Conda and R package set. No container digest is claimed because the scientific baseline was run in the recorded local environment rather than a container.

Acceptance ledger

The 52 plan requirements currently contain 43 verified, 6 partial and 3 open items. Partial entries name their remaining condition; they are not counted as complete.

Steps before publication

Order	Action	Evidence required
1	Independent quickstart	Fresh checkout, Python and operating-system versions, exact commit, commands, outcome and any documentation correction
2	Candidate freeze	Final inventory, clean local checks, pushed commit and green hosted workflow
3	Owner approval	Approval of the exact commit that will receive the v1.0.0 tag
4	GitHub release	Annotated v1.0.0 tag and a normal release titled OpenOmicsBench 1.0.0
5	Zenodo completion	Vivaan Patni as creator, exact version DOI, version 1.0.0, licences and relevant keywords
6	Post-upload check	Downloaded archive bytes and SHA-256 values match the certified release files

Known publication dependencies

The concept DOI is 10.5281/zenodo.22551734. The earlier 0.1.0.dev0 record is 10.5281/zenodo.22551735. The final version DOI must be entered only after Zenodo creates the 1.0.0 record. The author ORCID can be added to CITATION.cff and Zenodo metadata when supplied by Vivaan Patni.

Version 2 direction

Likely next work includes read-level objects, another assay with an equally explicit contract, and broader installation testing across operating systems. Published v1 evidence and the v1.0.0 tag should remain fixed; corrections should use a patch release.

Decision

The candidate has passed local and hosted preflight and is ready for the independent quickstart. It is not yet ready for the final public tag because the three publication-dependent records are still absent.